

NoSQL Unit 3

HBase

Introduction to HBase

1. In HBase, what is the main use of the SCAN operation? (1 Mark)

- A) To write a new row to the table.
 - B) To fetch a single row using its row key.
 - C) To iterate over rows in a table based on filter criteria.
 - D) To delete rows from a table.
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2. Which filter in HBase is suitable for selecting rows where a specific column's value exceeds 100? (1 Mark)

- A) FamilyFilter
 - B) SingleColumnValueFilter
 - C) ColumnPaginationFilter
 - D) RowFilter
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HBase vs RDBMS

3. Which of the following is NOT a common data model used in NoSQL databases? (1 Mark)

- a) Key-value store
 - b) Column-family store
 - c) Document store
 - d) Relational table store
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Hive

Introduction to Hive

4. What is the correct HiveQL command to define a new table schema in Hive? (1 Mark)

- A) CREATE SCHEMA TABLE
 - B) DEFINE TABLE
 - C) CREATE TABLE
 - D) MAKE DATABASE
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5. Choose the correct statement about Apache Hive: (1 Mark)

- A) Hive is optimized for transactional workloads.
 - B) Hive allows SQL-like queries on large-scale data stored in Hadoop.
 - C) Hive runs independently of the Hadoop ecosystem.
 - D) Hive stores data in key-value formats like NoSQL systems.
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Create / Alter / Drop Table

6. What is the result of executing the Hive query: SELECT COUNT(*) FROM departments; ? (1 Mark)

- A) A list of departments with employee counts
 - B) The total number of department names
 - C) A detailed record of all departments
 - D) The number of rows in the departments table
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HiveQL – Group By and Having

7. How would you calculate the average age of employees in each department using Hive? (1 Mark)

- A) SELECT AVG(age), department FROM employees;
 - B) SELECT department FROM employees WHERE AVG(age);
 - C) SELECT department, AVG(age) FROM employees GROUP BY department;
 - D) SELECT department, COUNT(age) FROM employees GROUP BY department;
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8. In HiveQL, which clause is used to filter groups after aggregation? (1 Mark)

- a) where
 - b) having
 - c) group by
 - d) order by
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9. In HiveQL, what is the main difference between JOIN and GROUP BY? (1 Mark)

- a) JOIN combines rows from two or more tables; GROUP BY aggregates rows based on column values
 - b) JOIN filters rows; GROUP BY sorts rows
 - c) JOIN sorts rows; GROUP BY combines rows
 - d) JOIN aggregates rows; GROUP BY filters rows
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10. Explain how GROUP BY and HAVING clauses are used in Hive. Write a query to get the departments having more than 5 students. (6 Marks)

11. What is the difference between the WHERE and HAVING clauses in Hive? Write a Hive query to find courses which is enrolled more than 10 students. (4 Marks)

12. Write a Hive query to find the total number of books written by each author. Display the author name and the total count of books. (4 Marks)

- Books (book_id, title, author_id)
 - Authors (author_id, author_name)
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HiveQL – Joins

13. Given two tables, students(student_id, student_name, class_id, marks) and classes(class_id, class_name), write a HiveQL query to join them on class_id and select each student's name, class name, and marks. (4 Marks)

HiveQL – Operators / Functions / Group By

14. How would you calculate the average age of employees in each department using Hive? (1 Mark)

- A) SELECT AVG(age), department FROM employees;
 - B) SELECT department FROM employees WHERE AVG(age);
 - C) SELECT department, AVG(age) FROM employees GROUP BY department;
 - D) SELECT department, COUNT(age) FROM employees GROUP BY department;
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Redis (Related NoSQL Topic under Unit III Context)

Redis Data Types

15. Which REDIS datatype is best suited to store a collection of unique values? (1 Mark)

- a) string
- b) list
- c) set

d) hash

16. Describe the various Redis data types. For each type, mention a use case where it would be most suitable. (6 Marks)

17. What is Redis? Describe its main data types — String, List, Set, Hash, and Sorted Set — and provide an example for each. (4 Marks)

Graph Databases / Neo4j (Included in CO4 / CO5 Questions)

Introduction to Graph Databases

18. What is a key feature of graph databases? (1 Mark)

- a) fixed schema
 - b) efficient storage and querying of relationships
 - c) store data only in tabular format
 - d) use of SQL for all queries
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19. Which is the best definition for the graph database? (1 Mark)

- A) A system that organizes data in collections and documents.
 - B) A system that stores information in tabular formats.
 - C) A database built on the concept of graph theory using nodes and edges.
 - D) A relational database system optimized for joins.
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20. What makes Neo4j distinct as a database system? (1 Mark)

- A) It uses JSON-based querying.
- B) It integrates natively with MySQL.

- C) It leverages graph structures and the Cypher query language.
 - D) It's primarily used for time-series analysis.
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Cypher Queries

21. In Cypher (Neo4j), which clause is used to find patterns in the graph? (1 Mark)

- A) MATCH
 - B) FIND
 - C) QUERY
 - D) SEARCH
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22. Why would a Neo4j node have both :Customer and :VIP labels? (1 Mark)

- A) To increase indexing performance by duplicating labels
 - B) To allow querying the node as either :Customer or :VIP
 - C) Because each node must have a minimum of two labels
 - D) To prevent the node from being matched in generic queries
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23. Which neo4j clause is used to remove properties or labels from nodes? (1 Mark)

- a) delete
 - b) remove
 - c) foreach
 - d) limit
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24. Explain the fundamental idea behind graph databases and highlight the key differences between graph databases and relational database systems. (6 Marks)

25. Write a Cypher query using Neo4j to add a node using the SET, SKIP, Aggregation clause. Explain the syntax and effect of the query. (6 Marks)

26. Write Cypher queries to create 5 Category nodes (Electronics, Books, Fashion, Techs, Toys), iterate over a list to create Product nodes with names, set a price property on each product, and connect them to categories using BELONGS_TO relationships. Then, find and count products whose names contain the letter "T" and finally return all products sorted by price in ascending order. (6 Marks)